# DysonSpherain: Route-Conditioned Candidate Admission for Long-Horizon Agent Memory Retrieval

DysonSpherain Project

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**Abstract**

Long-horizon agent memory systems often fail before generation: the relevant evidence is never admitted into the candidate pool. This paper studies candidate admission as a measurable failure layer in long-horizon memory retrieval and introduces DysonSpherain, a route-conditioned retrieval framework that combines dense-preserving safe fusion, route-specific candidate admission, and parent-to-segment anchoring. Across LongMemEval, LoCoMo, KnowMe-Bench, and CloneMem, DysonSpherain distinguishes final reranking failures from first-stage admission failures, showing strong coverage on dialogue-style memory tasks while exposing residual bottlenecks on profile-centric and non-conversational personal traces. To keep claims auditable, all reported results are validated through an artifact-backed protocol that detects fallback runs, blocks non-comparable historical artifacts, and ties tables to reproducible metric files.

## 1 Introduction

Long-horizon agent memory is often evaluated at the answer level, but many errors are already decided before generation begins. Benchmarks such as LongMemEval, LoCoMo, KnowMe-Bench, and CloneMem expose this issue across interactive dialogue, multi-session memory, person understanding, and AI-clone-style personal traces [14, 6, 15, 2]. A memory system may store the right evidence and still fail if the retrieval front end never admits that evidence into the candidate set. This failure layer is especially visible in long-running CLI and agent memory settings, where evidence is spread across sessions, people, timestamps, profile states, parent documents, and non-conversational traces.

Standard dense, sparse, or hybrid retrieval can fail in different ways under this setting [12, 3, 1]. Dense retrieval may find semantically plausible but temporally wrong evidence. Sparse retrieval may overfit exact words and miss paraphrased profile state. Hybrid retrieval may increase breadth while allowing noisy channels to push dense anchors out of the candidate set. Parent-level retrieval may find the correct session while failing to surface the gold child segment. We refer to these errors collectively as candidate admission failures.

Our goal is not to maximize a single leaderboard score, but to expose where long-horizon memory retrieval fails before generation and to provide guarded mechanisms that repair admission without destabilizing dense anchors. DysonSpherain implements this view as a route-conditioned candidate admission framework. A query route controls which retrieval channels are enabled, dense anchors are protected during fusion, and parent-level hits are expanded into bounded segment candidates before final reranking.

The paper makes four contributions. First, it defines candidate admission as a measurable failure layer for long-horizon memory retrieval. Second, it presents a route-conditioned admission framework with dense-preserving safe fusion and parent-to-segment anchoring. Third, it reports artifact-backed results across LongMemEval, LoCoMo, KnowMe-Bench, and CloneMem while preserving blocked and non-comparable evidence boundaries. Fourth, it turns low CloneMem performance into a failure-analysis contribution: CloneMem exposes residual admission bottlenecks over non-conversational, longitudinal personal traces rather than a solved success case.

# 2 Candidate Admission Failures in Long-Horizon Memory

Let $D = \{d_i\}$ be a memory corpus and $q$ be a query. Each query has a gold evidence set $G_q$ supplied by the benchmark. A first-stage retriever admits a candidate set $A_K(q) \subset D$ under budget $K$, and a final ranker returns $R_k(q)$ under output budget $k$. We write $r(q)$ for the route or query type, and $C_c(q)$ for the candidate set produced by channel $c$.

We measure first-stage admission with

$$\text{AdmissionRecall@K}(q) = \mathbf{1}[G_q \cap A_K(q) \neq \emptyset].$$

For segment-level benchmarks, we also track fractional evidence coverage:

$$\text{SegmentRecall@k}(q) = \frac{|G_q \cap R_k(q)|}{|G_q|}.$$

This distinction matters because a system can have high parent-level recall but low segment-level recall, or high candidate recall but poor final ordering.

We separate four failure layers. An *admission miss* occurs when $G_q \cap A_K(q) = \emptyset$. A *fusion or ordering failure* occurs when a gold item is admitted by some channel but is pushed below the candidate budget after fusion. A *reranking failure* occurs when the gold item reaches the rerank pool but does not survive into $R_k(q)$. A *parent-hit segment-miss* occurs when a parent/session containing the evidence is available but the correct child segment is not selected.

# 3 DysonSpherain Method

## 3.1 Problem Formulation

DysonSpherain builds $A_K(q)$ as a union of route-enabled channels and then applies fusion and parent-to-segment expansion:

$$A_K(q) = \text{Fuse}_K\left(P_{\text{dense}}(q), \{C_c(q) : c \in \mathcal{C}(r(q))\}\right).$$

Here $P_{\text{dense}}(q)$ is a protected dense anchor set and $\mathcal{C}(r(q))$ is the set of retrieval channels enabled by the route.

## 3.2 Route-Conditioned Candidate Admission

The implemented route taxonomy is deliberately conservative: semantic/dense, lexical/exact-anchor, temporal, entity/profile, and parent/session routes. A non-conversational trace route is not claimed as implemented general behavior; CloneMem suggests it as future work. The route classifier is a heuristic feature extractor over query text, detected entities, temporal cues, exact phrases, and benchmark route metadata:

$$r(q) = \text{RouteClassifier}(q).$$

The route enables a subset of channels and budgets:

$$A(q) = \bigcup_{c \in \mathcal{C}(r(q))} C_c(q).$$

Semantic routes fire when the query lacks explicit anchors and relies mainly on dense similarity. Lexical/code routes fire on quoted strings, names, identifiers, and exact phrases. Temporal routes fire on dates, durations, ordering language, and session-relative time. Entity/profile routes fire on person attributes, preferences, and state changes. Parent/session routes fire when the query is likely to require a larger episode before selecting a child segment. The currently available artifacts record full route-policy parameters only for some runs; missing route metadata is reported as `not recorded` rather than reconstructed by hand.

**Algorithm 1** DysonSpherain route-conditioned candidate admission

**Input:** query $q$, memory corpus $D$, candidate budget $K$, output budget $k$
$route \leftarrow classify\_route(q)$
$dense\_candidates \leftarrow dense\_retrieve(q, D)$
$route\_candidates \leftarrow retrieve\_by\_enabled\_routes(q, route, D)$
$protected \leftarrow top_m(dense\_candidates)$
$fused \leftarrow dense\_preserving\_fusion(protected, dense\_candidates, route\_candidates)$
$parents \leftarrow select\_parent\_anchors(fused)$
$segments \leftarrow parent\_to\_segment\_expand(parents, route)$
$expanded \leftarrow bounded\_union(fused, segments, K)$
$ranked \leftarrow rerank(expanded, route)$
$diagnostics \leftarrow attribute\_failures(q, route, dense\_candidates, fused, segments, ranked)$
**return** $top_k(ranked), diagnostics$

### 3.3 Dense-Preserving Safe Fusion

Fusion uses a channel-gated reciprocal-rank style score:

$$s_{\text{fuse}}(d, q) = \sum_{c \in \mathcal{C}(r(q))} \frac{w_c(r(q))}{k_0 + \text{rank}_c(d, q)}.$$

Dense protection keeps the top dense anchors from being displaced by noisy route-specific channels:

$$A_K(q) = P_{\text{dense}}(q) \cup \text{TopK}_{K-|P_{\text{dense}}|}\left(s_{\text{fuse}}(d, q)\right).$$

This is the mechanism that prevents broad channel fan-out from silently damaging LongMemEval and LoCoMo while still allowing lexical, temporal, profile, and parent/session evidence to enter when the route requires it. Duplicate collapse and parent diversity caps run after channel fusion so that one large parent or near-duplicate cluster cannot dominate the final candidate budget. Competition inhibition is treated as diagnostic and route-sensitive; it is not claimed as a universally enabled improvement.

### 3.4 Parent-to-Segment Anchoring

Many benchmark gold labels are segment-level even when retrieval first finds a larger parent document or session. After fusion, DysonSpherain expands a bounded number of parent/session hits into child segments. Child segments are rescored using local lexical overlap, route evidence, parent score, and segment budget. The expansion is bounded to avoid turning parent recall into an unbounded scan: only selected parents are expanded and only top child segments are returned to the reranker. The selector treats a parent-hit segment-miss as repaired only when at least one gold child segment is admitted into the candidate pool and survives into the final top-$k$ evidence context. Otherwise the diagnostic remains a segment-selection failure even if parent-level recall is high.

### 3.5 Failure Attribution

Diagnostics are emitted alongside retrieval. The failure taxonomy maps stored candidate diagnostics to admission miss, dense-hit-but-fusion-downrank, parent-hit-segment-miss, reranker-dropped-gold, duplicate/local-crowding, route miss, and invalid/fallback run. Each attribution is generated from artifacts: channel hit flags, channel gold ranks, fused gold rank, rerank gold rank, candidate recall summaries, and fallback metadata.

## 4 Artifact-Backed Evaluation Protocol

The formal protocol is a credibility layer, not the main contribution. It checks that official rows are non-fallback, complete, registry-backed, and not mixed with blocked or non-comparable artifacts. Historical runs are marked `non_comparable` when benchmark, run type, question scope, embedding, fallback status, config hash, dataset version, or route policy differs.

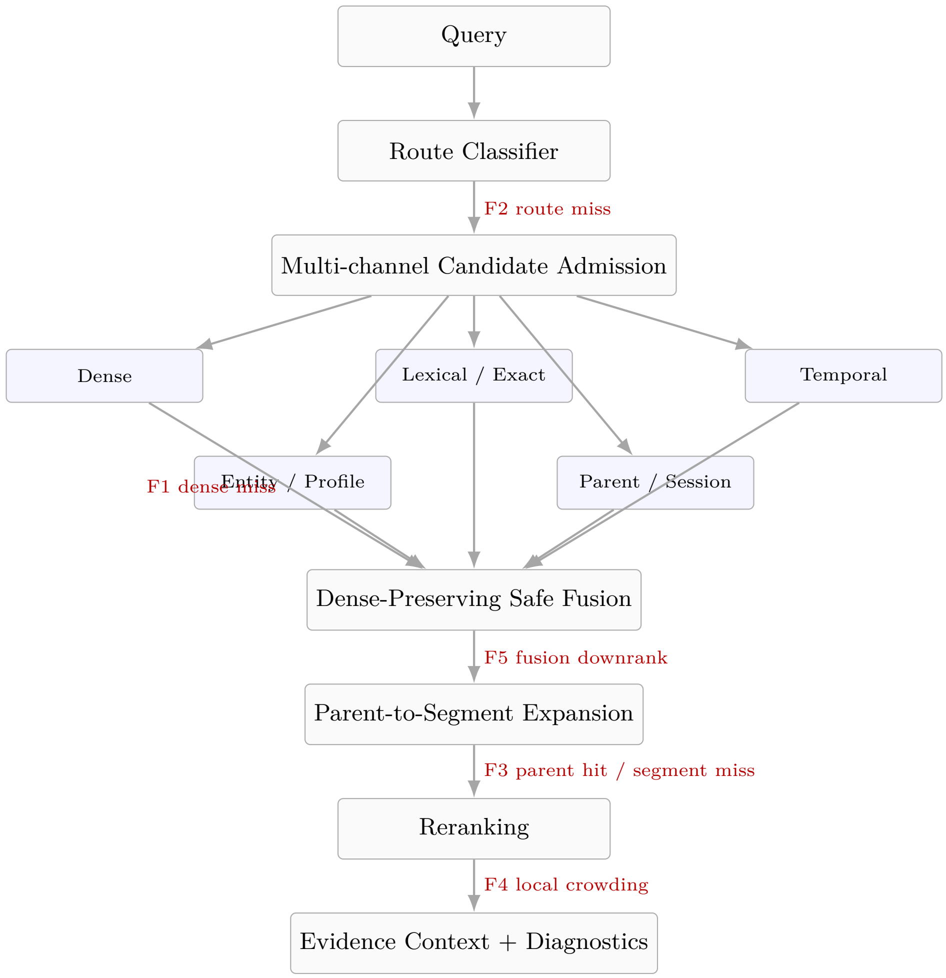


Figure 1: Method flow and failure points. The figure shows route classification, multi-channel admission, dense-preserving fusion, parent-to-segment expansion, reranking, and diagnostic attribution.

# 5 Experiments

## 5.1 Benchmarks

We evaluate LongMemEval, LoCoMo, KnowMe-Bench, and CloneMem. LongMemEval and LoCoMo stress long dialogue and multi-session memory. KnowMe-Bench stresses person understanding and lived-experience evidence. CloneMem stresses non-conversational personal traces and AI-clone-style memory continuity [14, 6, 15, 2].

## 5.2 Baselines

The main artifact table includes BM25, dense-only MiniLM, dense+BM25 RRF, and DysonSpherain full rows when available [12, 11, 1]. Blocked baselines such as unavailable cross-encoder/LLM rerankers and dense-only BGE/E5 rows are not converted into improvement claims.

## 5.3 Metrics

We report admission recall at 100, final recall or fractional recall at 10, NDCG at 10, elapsed seconds, and artifact status. For diagnostic sections, we also report channel hit flags and failure categories.

Table 1: Route-policy and budget metadata extracted from formal artifact metric files. Missing entries indicate that the formal artifact did not store the parameter, not that the value was zero.

| Parameter | LongMemEval | LoCoMo | KnowMe | CloneMem |
|---|---|---|---|---|
| dense_top_k | not recorded | not recorded | not recorded | 200 |
| protected_dense_top_m | not recorded | not recorded | not recorded | 100 |
| lexical_top_k | not recorded | not recorded | not recorded | 200 |
| temporal_top_k | not recorded | not recorded | not recorded | 90 |
| parent_top_k | not recorded | not recorded | not recorded | 50 |
| child_segment_cap | not recorded | not recorded | not recorded | 10 |
| final_k | not recorded | not recorded | not recorded | 100 |
| fusion | not recorded | not recorded | not recorded | rrf |
| duplicate_cap | not recorded | not recorded | not recorded | 1000 |
| route_policy_hash | not recorded | not recorded | not recorded | 9144917e6cd3c9e3eeef64bc709e02cbf2150b22b3 |

Table 2: Artifact validation coverage. Data source: `artifacts/formal_protocol_validation.json`.

| Evidence Section | Available | Blocked | Pending | Total |
|---|---|---|---|---|
| ablations | 36 | 0 | 0 | 36 |
| baselines | 28 | 8 | 0 | 36 |
| efficiency | 7 | 0 | 0 | 7 |
| leave_one_benchmark_out | 4 | 0 | 0 | 4 |
| statistics | 4 | 0 | 0 | 4 |

### 5.4 Implementation Details

All official full rows use the sentence-transformers MiniLM backend without local-hash fallback. CloneMem's protected top-3 lexical anchor gate is a route-only policy with route-policy hash `9144917e6cd3c9e3eeef64bc709e02cbf2150b2` It is not a global CLI default.

## 6 Results

### 6.1 Main Results

LongMemEval and LoCoMo show strong first-stage coverage with admission recall at 100 equal to 1.0. On these benchmarks, the remaining risk is less about broad candidate recall and more about fusion, duplicate collapse, ordering, and reranking. KnowMe-Bench exposes a different surface: admission recall is below LongMemEval/LoCoMo, suggesting profile, preference, entity, and temporal-object extraction remain incomplete. CloneMem remains a residual admission bottleneck and is reported as a stress test rather than a success case.

### 6.2 Ablations as Diagnostics

The ablation table should be read as diagnostic evidence, not as proof that every module is monotonically beneficial. Several ablated variants outperform the ablation-family full run on specific benchmarks. This is a finding rather than an inconsistency: admission mechanisms are surface-dependent. Lexical and temporal routes can recover anchors on one memory surface while adding noisy candidates on another; safe fusion can protect dense anchors while suppressing some sparse signals; parent expansion can repair parent-hit segment-miss cases while over-expanding local clusters. DysonSpherain is therefore a diagnostic and guarded route-conditioned framework, not a universal additive recipe.

Table 3: Formal full benchmark rows. Data source: `artifacts/formal_protocol_validation.json`. The admission-final gap is admission recall at 100 minus final recall at 10.

| Benchmark | System | Q | Admission R@100 | Final R@10 | Gap | NDCG@10 | Run ID |
|---|---|---|---|---|---|---|---|
| longmemeval | DysonSpherain formal full | 500 | 1.0000 | 0.9778 | 0.0222 | 0.9259 | longmemeval |
| locomo | DysonSpherain formal full | 1986 | 1.0000 | 0.9070 | 0.0930 | 0.7533 | locomo-a19d |
| knowme | DysonSpherain formal full | 1010 | 0.7266 | 0.5972 | 0.1294 | 0.5051 | knowme-641f |
| clonemem | DysonSpherain formal full | 2374 | 0.3442 | 0.0954 | 0.2488 | 0.0752 | clonemem-a4 |

Table 4: Available baseline rows. DysonSpherain rows are taken from the formal full source-of-truth artifact; baseline rows come from artifact-backed baseline tables. Missing Admission R@100 means the baseline artifact did not record candidate-recall diagnostics, so those rows are not used to support admission-specific claims.

| Benchmark | Method | Admission R@100 | Recall@10 | NDCG@10 | Elapsed s | Comparability |
|---|---|---|---|---|---|---|
| clonemem | bm25 | – | 0.0629 | 0.1022 | – | available; admissi |
| clonemem | dense_bm25_rrf | – | 0.0821 | 0.1265 | – | available; admissi |
| clonemem | dense_only_minilm | – | 0.0739 | 0.1094 | – | available; admissi |
| clonemem | dysonspherain_formal_full | 0.3442 | 0.0954 | 0.0752 | 3429.8 | non_comparable |
| knowme | bm25 | – | 0.4329 | 0.2862 | – | available; admissi |
| knowme | dense_bm25_rrf | – | 0.4168 | 0.3879 | – | available; admissi |
| knowme | dense_only_minilm | – | 0.5232 | 0.3656 | – | available; admissi |
| knowme | dysonspherain_formal_full | 0.7266 | 0.5972 | 0.5051 | 1872.8 | non_comparable |
| locomo | bm25 | – | 0.9059 | 0.7747 | – | available; admissi |
| locomo | dense_bm25_rrf | – | 0.9163 | 0.7170 | – | available; admissi |
| locomo | dense_only_minilm | – | 0.8340 | 0.6426 | – | available; admissi |
| locomo | dysonspherain_formal_full | 1.0000 | 0.9070 | 0.7533 | 528.6 | non_comparable |
| longmemeval | bm25 | – | 0.9620 | 0.8242 | – | available; admissi |
| longmemeval | dense_bm25_rrf | – | 0.9880 | 0.9089 | – | available; admissi |
| longmemeval | dense_only_minilm | – | 0.9880 | 0.8955 | – | available; admissi |
| longmemeval | dysonspherain_formal_full | 1.0000 | 0.9778 | 0.9259 | 496.2 | non_comparable |

### 6.3 Failure Analysis

CloneMem exposes the clearest residual failure. Candidate recall at 100 remains low and final recall/NDCG are also low. This points to first-stage admission over non-conversational, longitudinal, person-state traces rather than a pure final-reranking issue. The next mechanism frontier is a non-conversational trace route, stronger profile-state extraction, and temporal object extraction.

KnowMe-Bench also separates first-stage and final ordering failures. Its admission recall is lower than LongMemEval/LoCoMo, so the likely bottleneck is profile and preference evidence admission before final reranking.

### 6.4 Efficiency Analysis

Phase 13 provides seven matched budget artifacts for candidate and rerank budgets. Where matched quality metrics are unavailable for every budget, the paper keeps the efficiency plot descriptive and avoids inventing a Pareto gain.

The April 30 CloneMem efficiency run reduces full-run wall-clock time from the formal source-of-truth row's 3429.8 seconds to 1789.9 seconds on the same 2374 questions, with a 3.34× shard-level speedup estimate. This is an engineering result, not a retrieval-quality result: candidate recall at 100 is 0.3436 and final recall at 10 is 0.0950, matching the existing CloneMem bottleneck profile. The manuscript therefore updates the runtime claim while preserving the conclusion that CloneMem remains unsolved.

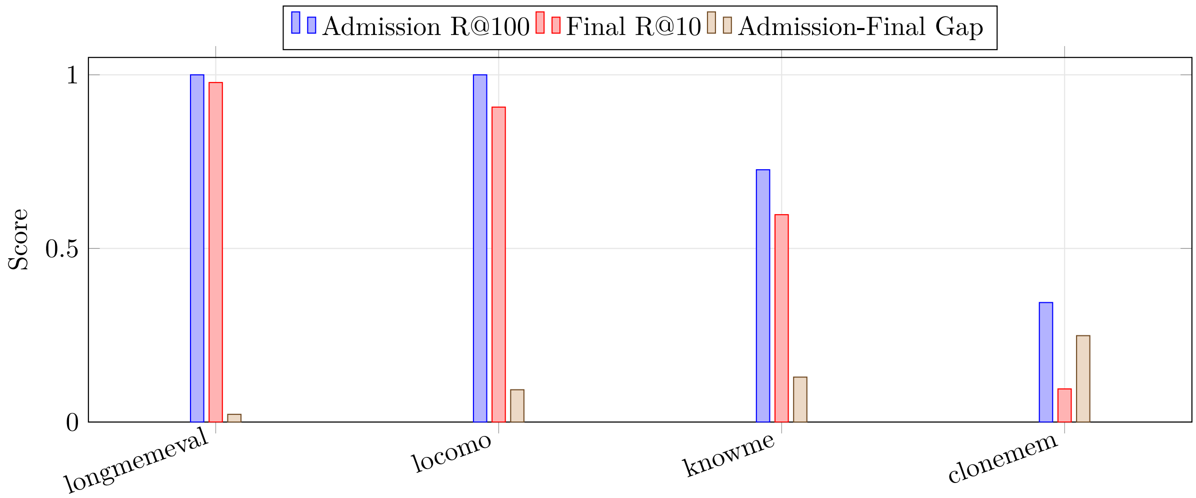


Figure 2: Admission-oriented benchmark view for formal full runs. The figure reports admission recall, final recall, and the admission-to-final gap rather than a generic benchmark score.

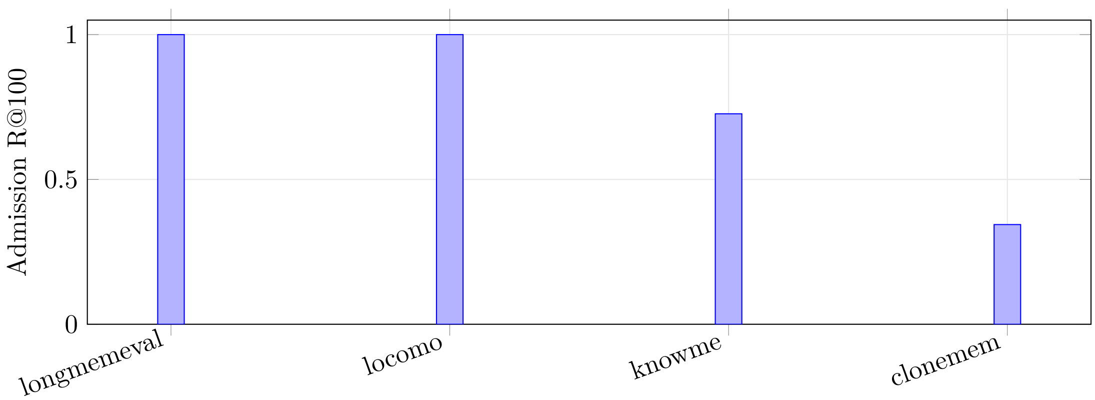


Figure 3: Admission recall by full formal benchmark. CloneMem and KnowMe show residual admission bottlenecks relative to LongMemEval and LoCoMo.

# 7 Case Studies

## 7.1 Diagnostic Case Studies

The examples below are extracted from diagnostic JSONL artifacts and redacted/truncated for paper use. They illustrate failure attribution rather than new benchmark claims.

**Case 1: Temporal anchoring drift.** Benchmark: `clonemem`. Failure type: `temporal_miss`. Admission R@100 for the query is 0.5; final R@10 is 0.0. Dense rank: 21; fused rank: 21; final rank: 55. Gold-hitting channels: dense_semantic, entity_aware, exact_phrase, lexical_sparse. Query excerpt: “[redacted]15[redacted]25[redacted]7[redacted]”

**Case 2: Parent-hit segment miss.** Benchmark: `clonemem`. Failure type: `parent_hit_segment_miss`. Admission R@100 for the query is 0.0; final R@10 is 0.0. Dense rank: 136; fused rank: –; final rank: –. Gold-hitting channels: dense_semantic, entity_aware, lexical_sparse. Query excerpt: “Old Chen, when you decided to delete that "Family Health Management Spreadsheet" back in early September, did you ever stop to think how things between you and Meifang...”

Table 5: Diagnostic ablation deltas generated from `paper/figures/data/ablation_waterfall_data.csv`. Deltas are relative to the ablation-family full row, not the formal full run. Positive values mean the ablated variant is higher on that metric.

| Benchmark | Removed Module | ΔAdmission R@100 | ΔFinal R@10 | ΔNDCG@10 | Interpretation |
|---|---|---|---|---|---|
| clonemem | -Route | 0.2308 | 0.0592 | 0.0497 | route policy sensitivity |
| clonemem | -Lexical | 0.2301 | 0.0754 | 0.0632 | lexical route can help anchors or |
| clonemem | -Temporal | 0.2306 | 0.0578 | 0.0535 | temporal route is surface-depend |
| clonemem | -Parent2Seg | 0.2306 | 0.0355 | 0.0306 | parent-to-segment selection risk |
| clonemem | -SafeFusion | 0.2432 | 0.0703 | 0.0505 | dense protection tradeoff |
| knowme | -Route | 0.0005 | -0.0010 | -0.0013 | route policy sensitivity |
| knowme | -Lexical | -0.0074 | 0.0104 | -0.0039 | lexical route can help anchors or |
| knowme | -Temporal | 0.0008 | -0.0037 | -0.0025 | temporal route is surface-depend |
| knowme | -Parent2Seg | -0.0037 | -0.0011 | -0.0039 | parent-to-segment selection risk |
| knowme | -SafeFusion | 0.0054 | -0.0133 | -0.0202 | dense protection tradeoff |
| locomo | -Route | 0.0000 | 0.0000 | 0.0000 | route policy sensitivity |
| locomo | -Lexical | 0.0000 | -0.0543 | -0.0885 | lexical route can help anchors or |
| locomo | -Temporal | 0.0000 | -0.0005 | -0.0008 | temporal route is surface-depend |
| locomo | -Parent2Seg | 0.0000 | 0.0015 | 0.0006 | parent-to-segment selection risk |
| locomo | -SafeFusion | 0.0000 | 0.0077 | 0.0098 | dense protection tradeoff |
| longmemeval | -Route | 0.0000 | 0.0000 | 0.0000 | route policy sensitivity |
| longmemeval | -Lexical | 0.0000 | -0.0046 | -0.0219 | lexical route can help anchors or |
| longmemeval | -Temporal | 0.0000 | 0.0037 | -0.0014 | temporal route is surface-depend |
| longmemeval | -Parent2Seg | 0.0000 | 0.0000 | 0.0000 | parent-to-segment selection risk |
| longmemeval | -SafeFusion | 0.0000 | 0.0047 | 0.0039 | dense protection tradeoff |

Table 6: CloneMem full-run efficiency update. The April 30 run is included as runtime evidence only: it uses the official embedding/vector path and covers the full CloneMem question set, but it does not change the paper's quality conclusion because candidate admission remains low.

| Run | Q | Wall s | Speedup | Admission R@100 | Final R@10 | Final NDCG@10 |
|---|---|---|---|---|---|---|
| Formal full | 2374 | 3429.8 | – | 0.3442 | 0.0954 | 0.0752 |
| Apr. 30 efficiency | 2374 | 1789.9 | 3.34 | 0.3436 | 0.0950 | 0.0749 |

The efficiency run is official-embedding and Chroma-backed, but it is used as runtime evidence; quality remains a warning because CloneMem candidate recall is still low. Serial shard time sums to 5983.4 seconds.

**Case 3: Reranker dropped gold.** Benchmark: `clonemem`. Failure type: `reranker_dropped_gold`. Admission R@100 for the query is 0.8; final R@10 is 0.0. Dense rank: 1; fused rank: 1; final rank: 12. Gold-hitting channels: dense_semantic, entity_aware, exact_phrase, lexical_sparse. Query excerpt: "Old Chen, I remember you mentioning that "stability" is your guiding principle, but over the past month or so, I feel like you[redacted]ve changed quite a bit. First,..."

**Case 4: KnowMe segment admission failure.** Benchmark: `knowme`. Failure type: `parent_hit_segment_miss`. Admission R@100 for the query is 0.0; final R@10 is 0.0. Dense rank: –; fused rank: –; final rank: –. Gold-hitting channels: lexical_sparse, temporal_neighbor. Query excerpt: "What are the names of the two rats who shot at Karl Larsdal and the narrator in the garbage dump?"

## 8 Related Work

**Long-horizon memory benchmarks.** LongMemEval and LoCoMo emphasize long dialogue and multi-session memory. KnowMe-Bench emphasizes person understanding and lived-experience evidence. Clone-

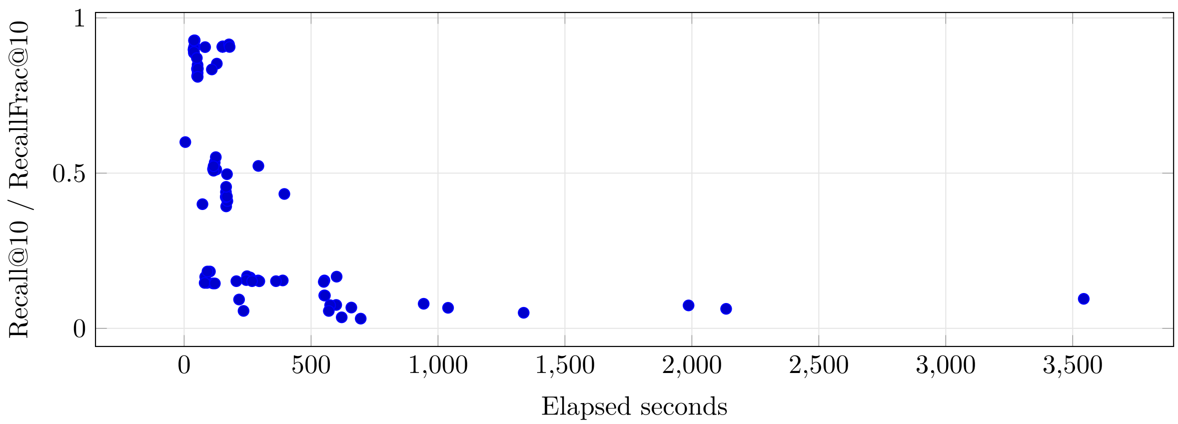


Figure 4: Efficiency-quality surface from available artifacts. Points mix available diagnostic and formal records, so the figure is descriptive rather than a matched improvement claim.

Table 7: Diagnostic case-study examples extracted from JSONL artifacts.

| Case | Dense Rank | Fused Rank | Parent Rank | Segment Rank | Final Rank | Fixed |
|---|---|---|---|---|---|---|
| 1. Temporal anchoring drift | 21 | 21 | – | 21 | 55 | not fi |
| 2. Parent-hit segment miss | 136 | – | – | – | – | not fi |
| 3. Reranker dropped gold | 1 | 1 | – | 1 | 12 | not fi |
| 4. KnowMe segment admission failure | – | – | – | – | – | not fi |

Mem emphasizes non-conversational personal traces and AI-clone-style continuity [14, 6, 15, 2]. Other agent-memory benchmarks cover adjacent surfaces including tool-use, life-assistant memory, and episodic or semantic memory evaluation. DysonSpherain differs by focusing on the admission layer before generation.

**Agent memory systems.** Systems such as Mem0, MemGPT/Letta, LangMem, MemoryBank, Generative Agents, Reflexion, and MemAgent study memory writing, updating, retrieval, reflection, and agent behavior [7, 9, 4, 16, 10, 13]. This paper is narrower: it treats candidate admission as a measurable retrieval failure layer and reports diagnostics that distinguish admission, fusion, parent-to-segment, and reranking failures.

**Retrieval and reranking.** Dense retrieval, BM25, hybrid retrieval, reciprocal rank fusion, cross-encoder reranking, parent-child retrieval, and multi-stage retrieval are standard tools for document retrieval and RAG [12, 3, 1, 8, 5]. DysonSpherain is not simply hybrid retrieval: it uses route-conditioned channel selection, dense-preserving fusion, bounded parent-to-segment expansion, and diagnostic attribution for long-horizon memory queries.

**Personalization and temporal memory.** Personal memory systems must reason over temporal context, entity-centric state, preferences, profile changes, and forgetting. These requirements make privacy and provenance central rather than peripheral.

## 9 Limitations

This paper does not claim universal state of the art. Historical runs marked `non_comparable` are not used for improvement claims. CloneMem remains a residual admission bottleneck. Ablations are not monotonic across benchmarks, which means route-policy selection remains a research problem. Baseline admission metrics still require full comparable reruns when candidate-recall diagnostics were not stored. Route policies carry benchmark-specific tuning risk and should not be treated as global defaults. The method optimizes retrieval and admission; it does not directly solve generation hallucination. The validation protocol blocks local embedding fallback from entering formal tables. Finally, benchmark gold granularity may not always match the system's memory segmentation, making parent-to-segment selection a continuing source of error.

# 10 Ethics and Privacy Considerations

Long-term personal memory can contain sensitive information about identity, preferences, health, emotions, routines, relationships, and behavior patterns. Systems should not infer sensitive attributes by default. Practical deployments need deletion, forgetting, redaction, provenance tracking, and retrieval traces. AI-clone scenarios can amplify privacy and identity-simulation risks, so benchmark performance should not be interpreted as sufficient proof of real user-modeling safety. Future CLI, email, calendar, or code-agent memory integrations should be opt-in, local-first where possible, and governed by user-controlled retention policies.

# 11 Conclusion

DysonSpherain reframes long-horizon agent memory retrieval around candidate admission failures. The method combines route-conditioned admission, dense-preserving safe fusion, and parent-to-segment anchoring, while the artifact protocol prevents fallback, blocked, and non-comparable evidence from becoming inflated claims. The current results show strong admission on LongMemEval and LoCoMo, expose profile and trace bottlenecks on KnowMe and CloneMem, and define a concrete next frontier for non-conversational personal memory retrieval.

# A Artifact Schema and Validation Protocol

Formal evidence is validated through artifact JSON, registry metadata, fallback flags, dataset and config hashes, route-policy hashes, and explicit blocked/non-comparable statuses. The summary table in Table 2 is generated from `artifacts/formal_protocol_validation.json`.

# B Full Benchmark Tables

The full paper-facing baseline table is generated by `scripts/generate_paper_tables.py`. Blocked unavailable-model rows remain in artifact reports but are excluded from main improvement claims.

# C Full Ablation Tables

Table 5 is generated from the ablation CSV and should be interpreted together with benchmark-specific comparability notes.

# D Failure Taxonomy

# E Additional Case Studies

Additional diagnostic examples can be regenerated with `scripts/generate_case_studies.py`.

## F Hyperparameters and Compute

Official full rows report elapsed seconds in Table 3. Route budgets and config hashes are tracked in the benchmark artifacts and registry. No cross-encoder or unavailable BGE/E5 rows are promoted into formal results.

## G NeurIPS Checklist

The current draft includes limitations, ethics, reproducibility notes, artifact-backed tables, and explicit blocked/non-comparable status handling.

## References


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Table 8: Failure taxonomy artifact rows.

| Benchmark | Ablation | Artifact Rows |
|---|---|---:|
| clonemem | full_admission | 1 |
| clonemem | no_benchmark_route_tuning | 1 |
| clonemem | no_inhibition | 1 |
| clonemem | no_lexical_probe | 1 |
| clonemem | no_parent_to_segment_selector | 1 |
| clonemem | no_rerank_guard | 1 |
| clonemem | no_route_conditioned_admission | 1 |
| clonemem | no_safe_fusion | 1 |
| clonemem | no_temporal_routing | 1 |
| knowme | full_admission | 1 |
| knowme | no_benchmark_route_tuning | 1 |
| knowme | no_inhibition | 1 |
| knowme | no_lexical_probe | 1 |
| knowme | no_parent_to_segment_selector | 1 |
| knowme | no_rerank_guard | 1 |
| knowme | no_route_conditioned_admission | 1 |
| knowme | no_safe_fusion | 1 |
| knowme | no_temporal_routing | 1 |
| locomo | full_admission | 1 |
| locomo | no_benchmark_route_tuning | 1 |
| locomo | no_inhibition | 1 |
| locomo | no_lexical_probe | 1 |
| locomo | no_parent_to_segment_selector | 1 |
| locomo | no_rerank_guard | 1 |
| locomo | no_route_conditioned_admission | 1 |
| locomo | no_safe_fusion | 1 |
| locomo | no_temporal_routing | 1 |
| longmemeval | full_admission | 1 |
| longmemeval | no_benchmark_route_tuning | 1 |
| longmemeval | no_inhibition | 1 |
| longmemeval | no_lexical_probe | 1 |
| longmemeval | no_parent_to_segment_selector | 1 |
| longmemeval | no_rerank_guard | 1 |
| longmemeval | no_route_conditioned_admission | 1 |
| longmemeval | no_safe_fusion | 1 |
| longmemeval | no_temporal_routing | 1 |